

Section : Chemistry

- Q.1** Out of the following given statements, the incorrect statements are;
 (A) 2, 2, 6-Trimethylcyclohexanone forms a cyanohydrin in good yield, but the cyclohexanone doesn't.
 (B) There are two amino groups in semicarbazide and both can be involved in the formation of semicarbazones.
 (C) Mononitration of benzaldehyde will form m-nitrobenzaldehyde.
 (D) Mononitration of acetophenone will form m-nitroacetophenone.
 Choose the correct answer from the options given below:
 1. (B), (C) and (D) only.
 2. (A), (B) and (C) only.
 3. (A), (B), (C) and (D).
 4. (A), (B) and (D) only

Options 1. 1

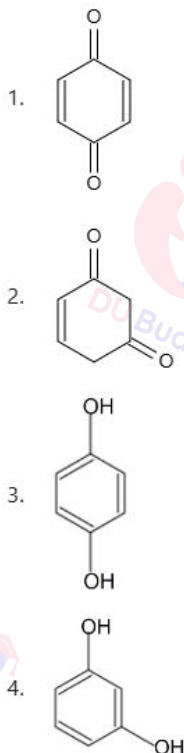
2. 2

3. 3

4. 4

Q.2

The correct structure for hydroquinone is:



Options 1. 1

2. 2

3. 3

4. 4

- Q.3** Arrange the following steps in the correct sequence to carry out reverse osmosis
 (A) Two solutions having different concentrations of the same solute are taken.
 (B) Very high external pressure is applied on the high concentration side.
 (C) A semipermeable membrane is inserted between the two solutions.
 (D) Solvent flows from the high concentration side to the low concentration one.
 Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (A), (C), (B), (D)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

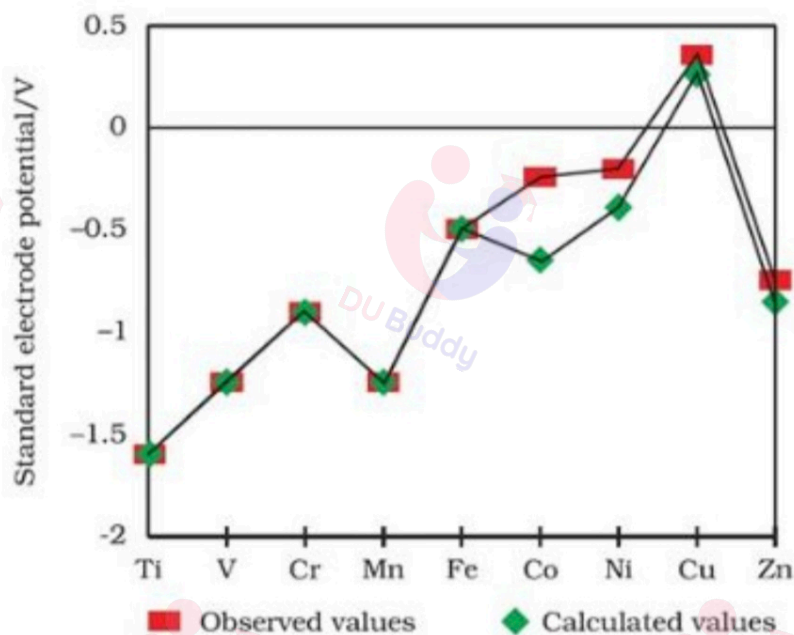
Options 1. 1

2. 2

3. 3

4. 4

Q.4



Observed and calculated values for the standard electrode potentials ($M^{2+} \rightarrow M$) are shown in the figure above. Based on the data shown, which of the following statements are true?

- (A) Cu is not able to liberate H_2 from acid.
 (B) Zn is not able to liberate H_2 from acid.
 (C) An increase in the sum of the first and second ionization enthalpies leads to less negative E° values.
 (D) Only oxidising acids react with Cu.

Choose the correct answer from the options given below:

- (A), (C) and (D) only
- (A), (B) and (C) only
- (A), (B) and (D) only
- (B), (C) and (D) only

Options 1. 1

- 2
- 3
- 4

Q.5 The rate constants of the reaction at 100 K and 200 K are 0.01 s^{-1} and 0.02 s^{-1} respectively.

The value of activation energy (E_a) will be

($\log 2 = 0.3010$)

- 576.33 J
- 576.33 J
- 1152.65 J
- 1152.65 J

Options 1. 1

- 2
- 3
- 4

Q.6 Match the biological entity given in List-I with the metal involved in it given in List-II

- | Biological entity | Metal involved |
|-------------------|----------------|
| (A) Chlorophyll | (I) Iron |
| (B) Haemoglobin | (II) Magnesium |
| (C) Vitamin B12 | (III) Nickel |
| (D) Urease | (IV) Cobalt |

Choose the correct answer from the options given below:

- (A) - (IV), (B) - (I), (C) - (III), (D) - (II)
- (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
- (A) - (III), (B) - (I), (C) - (IV), (D) - (III)
- (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Options 1. 1

- 2
- 3
- 4

Q.7 (a) $Ni(CO)_4$, (b) $[Ni(CN)_4]^{2-}$ and (c) $[Pt(NH_3)_2Cl_2]$ are respectively _____ complexes.

- All tetrahedral
- (a) Tetrahedral, (b) Square planar and (c) Tetrahedral
- (a) Square planar, (b) Tetrahedral, (c) Octahedral

4. (a) Tetrahedral, (b) Square planar and (c) Square planar

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.8 Arrange the following compounds in the increasing order of their boiling points:

- (A) Pentan-1-ol
(B) n-Butane
(C) Pentan-1-al
(D) Ethoxyethane

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (D), (C), (A)
4. (B), (C), (D), (A)

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.9 A solution contains 50 g of a non-electrolytic solute per 100g of water at 25°C. The vapor pressure of water is 23.5 mmHg at this temperature. The molar mass of the solute is 250 g mol⁻¹. The vapor pressure of the solution at 25°C will be

1. 20.6 mmHg
2. 22.67 mmHg
3. 24.00 mmHg
4. 19.60 mmHg

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.10 Arrange the following complexes in increasing order of number of moles of silver chloride precipitated when one mole of each is reacted with excess of silver nitrate.

- (A) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
(B) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
(C) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
(D) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

Choose the correct answer from the options given below:

1. (C), (A), (B), (D)
2. (D), (C), (B), (A)
3. (B), (A), (D), (C)
4. (A), (D), (B), (C)

- Options
1. 1
 2. 2
 3. 3
 4. 4

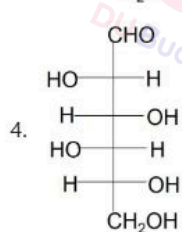
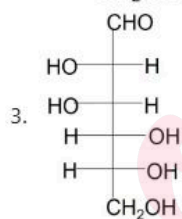
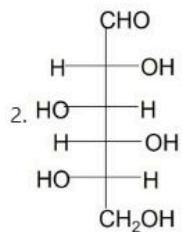
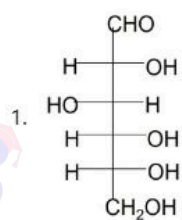
Q.11 Which of the following metal ions does not show color in aqueous medium?

1. Fe^{3+}
2. Cr^{3+}
3. V^{2+}
4. Ti^{4+}

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.12

The correct structure of D(+) Glucose is:



Options

1. 1
2. 2
3. 3
4. 4

Q.13 Lanthanoids (Ln) when heated with S, generally give _ _ _ _ _ _ _ _.

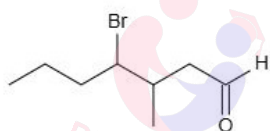
1. LnS_2
2. Ln_2S_3
3. Ln_2S
4. LnS_3

Options

1. 1
2. 2
3. 3
4. 4

Q.14

The correct IUPAC name of the following organic compound is



1. 4-Bromo-3-methylheptan-7-al
2. 3-Bromo-4-methylheptanal
3. 4-Bromo-3-methylheptanal
4. 3-Bromo-4-methylheptan-7-al

Options

1. 1
2. 2
3. 3
4. 4

Q.15 Which of the following compounds will show the least reactivity towards nucleophilic addition reaction?

1. Benzaldehyde
2. Acetophenone

3. p- Methylbenzaldehyde
4. p-Nitrobenzaldehyde

Options 1. 1
2. 2
3. 3
4. 4

- Q.16 A radioactive substance decays with a rate constant of $9.2 \times 10^{-3} \text{ min}^{-1}$ Time taken for 75% of the sample to decompose is
1. 150.7 min
2. 155.5 min
3. 65.5 min
4. 60.4 min

Options 1. 1
2. 2
3. 3
4. 4

- Q.17 The correct sequence for product formation according to the Collision theory of chemical reaction is
(A) Proper orientation
(B) Molecules collide with threshold energy
(C) Intermediate formation
(D) Formation of products
Choose the correct answer from the options given below:
1. (B), (A), (C), (D)
2. (A), (B), (D), (C)
3. (D), (A), (B), (C)
4. (B), (C), (A), (D)

Options 1. 1
2. 2
3. 3
4. 4

- Q.18 Match the complexes given in List-I with the number of unpaired electrons given in List-II

List-I List-II
Complexes No. Of unpaired electrons.

(A) $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ (I) 1

(B) $[\text{NiF}_6]^{4-}$ (II) 5

(C) $[\text{Fe}(\text{CN})_6]^{4-}$ (III) 2

(D) $[\text{MnF}_6]^{4-}$ (IV) 0

Choose the correct answer from the options given below:

1. (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
2. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
3. (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
4. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Options 1. 1
2. 2
3. 3
4. 4

- Q.19 The products for the reaction of KMnO_4 with potassium iodide in acidic solutions are]

1. Mn^{2+} , H_2O , I_2
2. Mn^{4+} , H^+ , IO_3^-
3. Mn^{4+} , H_2O , I_2
4. Mn^{2+} , H^+ , IO_3^-

Options 1. 1
2. 2
3. 3
4. 4

- Q.20 Addison's disease is not characterised by

1. Hypoglycemia
2. Weakness in the body.
3. Increased susceptibility to stress.
4. Modulated inflammatory reactions.

Options 1. 1
2. 2
3. 3
4. 4

- Q.21 Arrange the following compounds in the decreasing order of reactivity towards $\text{S}_\text{N}1$ reaction

- (A) $(\text{Ph})_3\text{C-I}$
(B) $(\text{Ph})_3\text{C-Br}$
(C) $(\text{Ph})_3\text{C-Cl}$
(D) $(\text{Ph})_3\text{C-F}$

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (B), (D), (C)
3. (D), (C), (B), (A)
4. (B), (A), (D), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Match List-I with List-II

List-I Solute	List-II Van't Hoff factor (i)
(A) Benzoic acid in benzene	(I) 3
(B) K_2SO_4 (aq)	(II) 2
(C) Urea (aq)	(III) 0.5
(D) KOH (aq)	(IV) 1

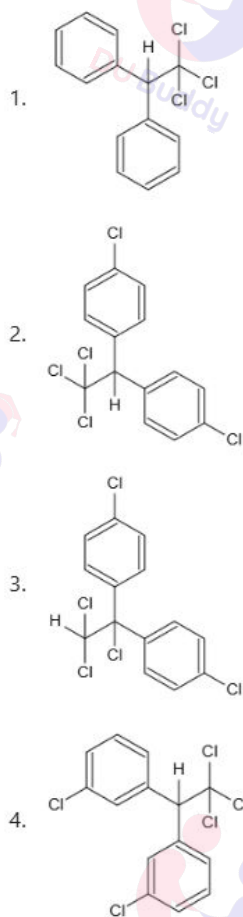
Choose the correct answer from the options given below:

1. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
2. (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
3. (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
4. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.23

The correct structure of DDT is;



- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.24 To avoid the medical condition called 'bends' in scuba divers, the composition of gas filled in the tank is

1. Helium 11.7%, Nitrogen 56.2%, Oxygen 32.1%
2. Helium 1.7%, Nitrogen 63.2%, Oxygen 35.1%
3. Helium 2.5%, Nitrogen 60.5%, Oxygen 37%
4. Helium 2%, Nitrogen 78%, Oxygen 22%

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.25 Select the high spin complexes from the following:

- (A) $[\text{Ni}(\text{CN})_4]^{2-}$
(B) $[\text{Fe}(\text{CN})_6]^{3-}$
(C) $[\text{MnBr}_4]^{2-}$
(D) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

Choose the correct answer from the options given below:

1. (B) and (C) only
2. (C) and (D) only
3. (A) and (D) only
4. (A) and (B) only

Options 1. 1
2. 2
3. 3
4. 4

Q.26 With reference to the elimination reaction of alkyl halides, the correct statements are;

- (A) Zaitsev rule predicts the major product in dehydrohalogenation reaction.
(B) Grignard's reagent is prepared under perfectly anhydrous conditions.
(C) Alkyl halides on reaction with aqueous KOH forms an eliminated product but with alcoholic KOH forms a substituted product.
(D) Alkyl halides on reaction with aqueous KOH forms a substituted product but with alcoholic KOH forms an eliminated product.

Choose the correct answer from the options given below:

1. (A), (B), (C) and (D)
2. (B), (C) and (D) only
3. (A), (B) and (D) only.
4. (A), (B) and (C) only.

Options 1. 1
2. 2
3. 3
4. 4

Q.27 When a protein in its native form, is subjected to physical or chemical change, then its

1. The primary and secondary structures remain intact.
2. The secondary and tertiary structures remain intact.
3. Only the secondary structure remains intact.
4. Only the primary structure remains intact.

Options 1. 1
2. 2
3. 3
4. 4

Q.28 Which of the following is the best reducing agent to convert RCOOH into RCH_2OH ?

1. LiBH_4
2. B_2H_6
3. NaBH_4
4. Hydrogen in presence of Pd

Options 1. 1
2. 2
3. 3
4. 4

Q.29 The major product of Kolbe's reaction is used for the synthesis of a precursor substrate in the preparation of aspirin.

The electrophile used in Kolbe's reaction is

1. Carbon dioxide
2. Carbon monoxide
3. Acetic anhydride
4. Phenol

Options 1. 1
2. 2
3. 3
4. 4

Q.30 Match List-I with List-II

List-I	List-II
Name of Compound	Application
(A) Neutral Ferric Chloride	(I) Anaesthetic
(B) Methanol	(II) Detection of Phenolic group
(C) Diethylether	(III) Wood Spirit
(D) Acetylsalicylic acid	(IV) Analgesic

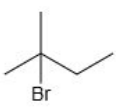
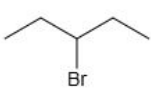
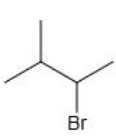
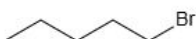
Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1
2. 2
3. 3
4. 4

Q.31

Which compound will undergo fastest S_N1 reaction?

1. 
2. 
3. 
4. 

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.32 Match List-I (type of linkages) with List-II (Biomolecules)

List-I	List-II
Type of linkage	Biomolecule
(A) Glycosidic linkage	(I) Tertiary structure of proteins
(B) Peptide Linkage	(II) Disaccharide
(C) Phosphodiester linkage	(III) Primary structure of proteins
(D) Disulfide linkage	(IV) Nucleotide

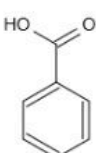
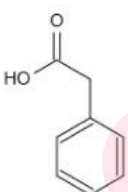
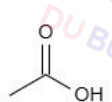
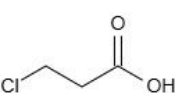
Choose the correct answer from the options given below:

1. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.33

Which of the following will have the highest pK_a value;

1. 
2. 
3. 
4. 

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.34 For a bimolecular elementary reaction between A and B, the rate of reaction is expressed as
 $\text{rate} = P \cdot Z_{AB} e^{-E_a/RT}$

Identify the terms P , Z_{AB} and E_a involved in this expression, respectively

1. Probability factor, collision number, Absolute energy
2. Pressure, collision number, Activation energy
3. Probability factor, collision frequency, Activation energy
4. Pressure, collision frequency, Absolute energy

Options 1. 1

2. 2

3. 3

4. 4

Q.35 For first order decomposition of N_2O_5 at constant volume, p_i is initial pressure and p_t is pressure at any time t during the reaction.

Which of the following statement/s is incorrect?

- (A) Half life does not depend on initial pressure
- (B) Total pressure decreases with time
- (C) Rate constant will be the same if initial pressure is doubled
- (D) $k = \frac{2.303}{t} \log \left(\frac{p_i}{p_t} \right)$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (D) only
3. (C) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.36 When a nucleotide from DNA containing adenine is hydrolyzed, the products obtained are

1. Adenine, deoxyribose sugar and phosphoric acid.
2. Adenine, ribose sugar and phosphoric acid
3. Adenine and ribose sugar only.
4. Adenine and deoxyribose sugar only.

Options 1. 1

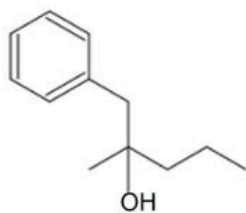
2. 2

3. 3

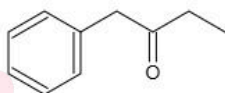
4. 4

Q.37

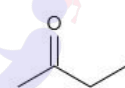
The substrates required for the preparation of the following alcohol are



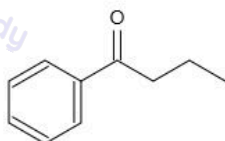
1. $\text{CH}_3\text{-MgBr}$ and



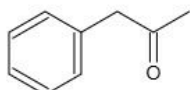
2. $\text{C}_6\text{H}_5\text{-MgBr}$ and



3. $\text{CH}_3\text{-MgBr}$ and



4. $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{MgBr}$ and



- Options
- 1
 - 2
 - 3
 - 4

Q.38 1.00 g of a non-electrolyte solute when dissolved in 50 g of organic solvent, lowered the freezing point of the organic solvent by 0.50 K.
If the freezing point depression constant of the organic solvent is 5 K kg mol^{-1} , the molar mass of the solute will be

1. 250 g mol^{-1}
2. 200 g mol^{-1}
3. 350 g mol^{-1}
4. 300 g mol^{-1}

- Options
- 1
 - 2
 - 3
 - 4

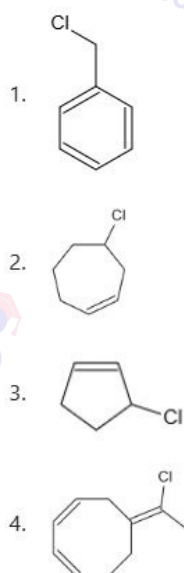
Q.39 Which compound of transition metal is used as a catalyst in the Contact Process?

1. Copper (II) oxide
2. Manganese (IV) oxide
3. Iron (III) oxide
4. Vanadium (V) oxide

- Options
- 1
 - 2
 - 3
 - 4

Q.40

Identify the allyl chloride from the following:



- Options
- 1
 - 2
 - 3
 - 4

Q.41 

Point C in the image is called

1. Conductivity limit
2. Limited equivalent conductance at infinite dilution
3. Limiting conductance at infinite dilution
4. Limiting molar conductivity

- Options
- 1
 - 2
 - 3
 - 4

Q.42 

The decrease in molar conductivity of a weak electrolyte on increasing the concentration for a fixed volume is due to

1. Increase in degree of dissociation on dilution
2. Decrease in degree of dissociation on dilution
3. Decrease in dissociation constant on increasing the concentration
4. Increase in dissociation constant on increasing the concentration

- Options
- 1
 - 2
 - 3

Q.43



Units of conductivity and molar conductivity are, respectively

1. S/cm , $\text{S cm}^2 \text{mol}^{-1}$
2. S/cm^2 , S cm mol^{-1}
3. S cm , $\text{S cm}^2 \text{mol}^{-1}$
4. S cm^2 , $\text{S cm}^{-2} \text{mol}^{-1}$

Options

1. 1
2. 2
3. 3
4. 4

Q.44



Identify electrolyte A and B in the given image.

1. A is a strong electrolyte, B is weak electrolyte.
2. A is a weak electrolyte, B is a strong electrolyte.
3. Both A and B are strong electrolytes with B having low solubility.
4. Both A and B are weak electrolytes.

Options

1. 1
2. 2
3. 3
4. 4

Q.45



A 0.001 mol L^{-1} solution of acetic acid shows conductivity of $5.02 \times 10^{-5} \text{ S cm}^{-1}$.

If Λ_m° for acetic acid is $390 \text{ cm}^2 \text{mol}^{-1}$, then the degree of dissociation is

1. 0.12
2. 0.50
3. 0.12×10^{-3}
4. 0.012

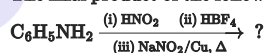
Options

1. 1
2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer

Organic compounds derived from NH_3 by replacing one or more H-atoms with alkyl or aryl groups will form primary, secondary and tertiary amines. The common methods which include their preparations are reduction of nitro group, amide group and nitrile group, Gabriel phthalimide synthesis and ammonolysis of alkyl halides. Amines are basic in nature due to the presence of a lone pair of electrons on the nitrogen atom. Important chemical properties of amines include acylation, alkylation and their interaction with nitrous acid. There are various tests which can distinguish different types of amines from each other, for e.g. carbylamine test, Hinsberg's test, coupling reaction etc. Another important class of compounds which can be derived from primary amines is azo compounds. These compounds are intensely colored and widely used as dyes in textile, leather and food industries.

The final product of the following reaction is:



1. Fluorobenzene
2. Nitrobenzene
3. 1-Fluoro-4-nitrobenzene
4. 4-Fluoro-1-nitrobenzene

Options

1. 1
2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer

Organic compounds derived from NH_3 by replacing one or more H-atoms with alkyl or aryl groups will form primary, secondary and tertiary amines. The common methods which include their preparations are reduction of nitro group, amide group and nitrile group, Gabriel phthalimide synthesis and ammonolysis of alkyl halides. Amines are basic in nature due to the presence of a lone pair of electrons on the nitrogen atom. Important chemical properties of amines include acylation, alkylation and their interaction with nitrous acid. There are various tests which can distinguish different types of amines from each other, for e.g. carbylamine test, Hinsberg's test, coupling reaction etc. Another important class of compounds which can be derived from primary amines is azo compounds. These compounds are intensely colored and widely used as dyes in textile, leather and food industries.

How many different structural isomers that correspond to the molecular formula $\text{C}_3\text{H}_9\text{N}$ are possible? Out of these, how many will give a positive carbylamine test?

1. 4 and 1
2. 3 and 1
3. 4 and 2
4. 3 and 2

Options

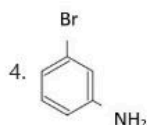
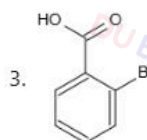
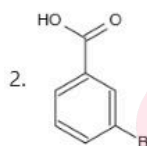
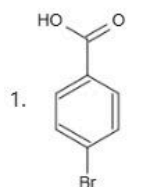
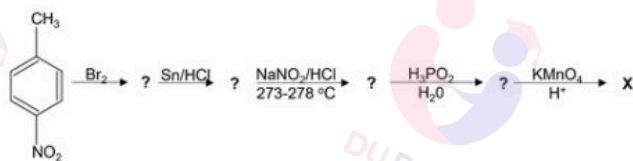
1. 1
2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer

Organic compounds derived from NH_3 by replacing one or more H-atoms with alkyl or aryl groups will form primary, secondary and tertiary amines. The common methods which include their preparations are reduction of nitro group, amide group and nitrile group, Gabriel phthalimide synthesis and ammonolysis of alkyl halides. Amines are basic in nature due to the presence of a lone pair of electrons on the nitrogen atom. Important

chemical properties of amines include acylation, alkylation and their interaction with nitrous acid. There are various tests which can distinguish different types of amines from each other, for e.g. carbylamine test, Hinsberg's test, coupling reaction etc. Another important class of compounds which can be derived from primary amines is azo compounds. These compounds are intensely colored and widely used as dyes in textile, leather and food industries.

In the following sequence of the conversion, identify the final product (X)



Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the passage carefully and answer

Organic compounds derived from NH_3 by replacing one or more H-atoms with alkyl or aryl groups will form primary, secondary and tertiary amines. The common methods which include their preparations are reduction of nitro group, amide group and nitrile group, Gabriel phthalimide synthesis and ammonolysis of alkyl halides. Amines are basic in nature due to the presence of a lone pair of electrons on the nitrogen atom. Important chemical properties of amines include acylation, alkylation and their interaction with nitrous acid. There are various tests which can distinguish different types of amines from each other, for e.g. carbylamine test, Hinsberg's test, coupling reaction etc. Another important class of compounds which can be derived from primary amines is azo compounds. These compounds are intensely colored and widely used as dyes in textile, leather and food industries.

Conversion of nitro group into primary amine is preferred using

1. Fe and HCl

2. Sn and HCl

3. Pt- H_2

4. Pd- H_2

Options 1. 1

2. 2

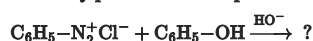
3. 3

4. 4

Q.50 Read the passage carefully and answer

Organic compounds derived from NH_3 by replacing one or more H-atoms with alkyl or aryl groups will form primary, secondary and tertiary amines. The common methods which include their preparations are reduction of nitro group, amide group and nitrile group, Gabriel phthalimide synthesis and ammonolysis of alkyl halides. Amines are basic in nature due to the presence of a lone pair of electrons on the nitrogen atom. Important chemical properties of amines include acylation, alkylation and their interaction with nitrous acid. There are various tests which can distinguish different types of amines from each other, for e.g. carbylamine test, Hinsberg's test, coupling reaction etc. Another important class of compounds which can be derived from primary amines is azo compounds. These compounds are intensely colored and widely used as dyes in textile, leather and food industries.

Correctly predict the final product and the type of reaction for the following reaction:



1. p-Aminoazobenzene, Nucleophilic Substitution Reaction

2. p-Aminoazobenzene, Electrophilic Substitution Reaction

3. *p*-Hydroxyazobenzene, Nucleophilic Substitution Reaction

4. *p*-Hydroxyazobenzene, Electrophilic Substitution Reaction

Options

1. 1

2. 2

3. 3

4. 4

